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Nesting and scheduling optimization of additive manufacturing systems: Mapping the territory

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ABSTRACT

The application of Additive Manufacturing (AM) within industrial settings brings new challenges to operational research. This paper provides in-depth analyses of nesting and scheduling problems in the context of AM. Using bibliometric and systematic review methods, we map the progression of knowledge in this territory and provide a detailed characterization of the existing mathematical models and approaches. Our findings indicate the fundamental decisions that predominate in these models and reveal the relationships between the critical studies, which form the backbone of the research. While the nesting and scheduling problem can be categorized into six types of decision, we have also found other emerging and interrelated optimization problems associated with nesting and scheduling decisions. To provide further details, an ample number of models are systematized in a reference framework for modeling nesting and scheduling in AM. Finally, pressing future research avenues that have emerged from our analysis and that need to be addressed to further advance the field are presented.

1. Introduction

Additive Manufacturing (AM) emerged out of rapid prototyping techniques and is often referred to as 3D printing (Attaran, 2017). In contrast to conventional subtractive and formative processing techniques, AM deposits successive layers of material, which enables the production of highly complex parts and offers new degrees of flexibility for product design (Li et al., 2019). Nevertheless, the introduction of AM into manufacturing systems is still in its infancy, and existing mathematical models still lag behind the practical needs of the new processes offered by AM. Therefore, at a time when consumers are calling for rapid responses, practical and efficient solutions rely on operational research and optimization models.

The scheduling problem is one of the most studied in operational research; nevertheless, AM brings additional challenges that have not received much attention until the last decade (Fera et al., 2020). This problem consists of grouping jobs into batches, and then schedule those batches over the available machines (Guzman et al., 2022) – referred as Batch Processing Machine (BPM) problem. AM machine scheduling is a variation of the pure BPM problem (Zhang et al., 2020). In this context, multiple parts, that correspond to jobs, are produced simultaneously in the same AM machine and during a single production cycle. Yet, in BPM

scheduling problems most of the existing research was conducted in the context of high-volume production and modeled as one-dimensional knapsack problems (Oh et al., 2020), which means that the machine capacity is constrained to a value, such as maximum weight, area, or volume. In AM scheduling problems, representing machine capacity involves modeling it as a nesting problem.

The nesting problem consists of packing multiple parts into a constrained build area or volume, which correspond to a batch in the BPM problem, for each production cycle (Alicastro et al., 2021). Typical decisions include the location and orientation of the parts within the build – this introduces more complexity into the primary AM scheduling problem. In addition, the AM machine capacity in these problems can be modeled as a two-dimensional (2D) or three-dimensional (3D) bin-packing problem. This is because certain processes, including Fused Deposition Modeling (FDM) and Stereolithography (SLA) require support structures, and it is not advisable to stack parts on top of each other. While other processes, such as Selective Laser Sintering (SLS), do not need support structures, and the machine capacity can be modeled as 3D. Therefore, the AM scheduling problem is characterized by integrating nesting constraints into BPM scheduling problem.

Both scheduling and nesting problems have been described in a handful of noteworthy reviews in recent years. Dyckhoff (1990)

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introduced the typology of Cutting and Packing (C&P) problems, which is closely related to the nesting problem. The author suggested four classification criteria whereby C&P problems can be arranged: dimensionality; kind of assignment; assortment of large objects; and assortment of small items. Wäscher et al. (2007) proposed another typology, which is partially based on Dyckhoff's original ideas. Araújo et al. (2019) focused on extending the existing typologies to the AM domain, proposing a classification according to four-tuples: dimensionality; optimization criteria; build volume type; and attributes of the assortment of parts. This paper was instrumental in bringing together nesting and AM problems. For scheduling problems, Graves (1981) introduced a classification with three different criteria: requirements generation; processing complexity; and scheduling criteria. These criteria have been updated and outlined into systematic taxonomies (Kanet and Sridharan, 2000; Potts and Kovalyov, 2000; Slotnick, 2011; Perez-Gonzalez and Framinan, 2014).

Recent reviews have started to integrate nesting and scheduling problems, departing from the traditional perspective of separately addressing these decisions. Oh et al. (2020) introduced a noteworthy taxonomy based on three entities: *Part*; *Build*; and *Machine*. Herein, the interaction from *Part* to *Build* and *Build* to *Machine* imply nesting and scheduling problems, respectively. Nevertheless, this classification is restricted to the root of the nesting and scheduling problem and does not delve into the immediate problems of AM, including part decomposition, part distribution, and pre- and post-processing operations. In addition, the authors did not extend the study to new AM scenarios, such as collaborative manufacturing and multi-robot manufacturing. Our study therefore provides an up-to-date review of how these new concerns are included within the nesting and scheduling problem.

De Antón et al. (2022) focused at the subproblem level. This level refers to breaking down the original nesting and scheduling problem into planning subproblems. Based on the hierarchy consisting of *Part*, *Build*, and *Machine*, they propose four planning stages to arrange the subproblems: *Part*; *Part-Machine*; *Part-Build*; and *Build-Machine*. The definitions of these subproblems were based on the general subproblems of Wäscher et al. (2007) and Maravelias and Sung (2009) and adapted to the AM context. While these authors focused on the subproblems at the operational level, we also study the integration of the operational with tactical level in AM.

Finally, Framinan et al. (2023) provide a literature review on the use of operations research methods in AM. The authors categorize AM decision problems based on three business functions: *Designing the AM process*; *Planning in an AM process*; and *Operating of AM*. Nevertheless, the decision problems were addressed separately, and there is no information of how these decisions are interrelated. This highlights the need for a review to consolidate the research on the integration of these AM decision problems, and to provide a coordination scheme for the different decisions.

Our review differs from the existing literature by specifically focusing on the incorporation of various decisions and constraints into the nesting and scheduling problem. Considering that these decisions are traditionally kept separate or loosely integrated, studying these interdependencies aims at contributing to the development of new models and streamline the decision-making process. To study the progression of the knowledge in this field, we present an analysis of the critical papers, which form the backbone of the research, and utilize a scheme according to the three recognized entities in AM: *Part*, *Build* and *Machine*. Furthermore, this study applies a more comprehensive procedure than a traditional review, by using science mapping techniques to capture and map the existing knowledge on nesting and scheduling problems in AM. Two research questions are answered:

RQ1. What is the research structure of nesting and scheduling problems in the context of AM?

RQ2. What are the characteristics prevalent in the modeling of nesting and scheduling problems in AM?

Through RQ1, this paper aims to systematize and rationalize the

previous research. Thus, we explore the intersection of qualitative and quantitative methods to ensure methodological rigor (Zupic and Čater, 2015). This produces the spatial representation of the field. We used two well-accepted bibliometric methods – conceptual analysis (Aria and Cuccurullo, 2017) and main path analysis (De Nooy et al., 2018). Specifically, conceptual analysis connects the fundamental concepts of the research field (Zupic and Čater, 2015). Afterwards, the main path analysis establishes the connections between the critical papers, which form the research backbone of the field (De Nooy et al., 2018). RQ2 navigates the characteristics of the existing mathematical models and approaches to nesting and scheduling problems according to the three entities: *Part*; *Build* and *Machine*; each one subject to different assumptions, which differ from model to model. Assumptions are referred to as the modeling gap, which represents the deviation between the problem characteristics and their representation in the model. The existing models are systematized in a reference framework for modeling nesting and scheduling in AM. Finally, a research program that summarizes pressing future research avenues that emerged from our analysis and that need to be addressed to further advance the field is presented.

The remainder of this paper is organized as follows. Section 2 outlines the methodology used. Section 3 presents the conceptual analysis and main path analysis. In Section 4, the characteristics of the existing mathematical models and approaches are analyzed, and the results systematized in a framework prior to Section 5, which outlines our future research program. Finally, Section 6 concludes.

2. Methodological approach

This paper follows the science mapping methodology proposed by Zupic and Čater (2015), which is comprised of five steps: research planning, data collection, data analysis, results visualization, and interpretation. All five are illustrated in Fig. 1. Web of Science (WoS) and Scopus were used for sourcing because they are widely used bibliographic databases. A set of keywords that comprehensively represents the sphere of interest was selected and combined into the following search string: (“*additive manufactur**” OR “*3D print**” OR “*rapid prototyp**”) AND (“*production planning*” OR “*scheduling*” OR “*nesting*” OR “*packing*”). The initial search was restricted to English-language peer-reviewed papers. For this initial search on December 18 of 2023 we retrieved a total of 963 papers from WoS and 1,324 papers from Scopus. Next, title and abstract were screened to confirm the relevance of the papers and duplicates were excluded. This reduced our sample to 143 papers. In a second screening process, where the full texts were read, some further papers were discarded. Our final sample consists of 130 papers: 105 journal articles and 25 conference proceedings. Finally, 7 articles were added through the snowball method.

The sample was subsequently exported to bibliometric software for further analysis. In this paper, the bibliometric analysis was primary conducted using Bibliometrix software, which is programmed in R language for statistical analyses (Aria and Cuccurullo, 2017), more specifically using the Biblioshiny interface. Pajek software, which has been extensively used to analyze network data, was also used to obtain the main path (De Nooy et al., 2018). Review papers were solely included for sample statistics, and for the conceptual and main path analysis. They were obviously excluded from the study on the characteristics of the models.

2.1. Terminology

Since the existing literature lacks a precise definition of the terminology, we use a set of essential terms to reference equivalent context-related terms in a bid to avoid confusion. For example, the terms referring to *additive manufacturing* have moved from *rapid prototyping* (see Gogate and Pande (2008), for example) to *3D printing* and *additive manufacturing* over time (see Fig. 2).

Similar to *additive manufacturing*, also for *nesting* and *scheduling* a

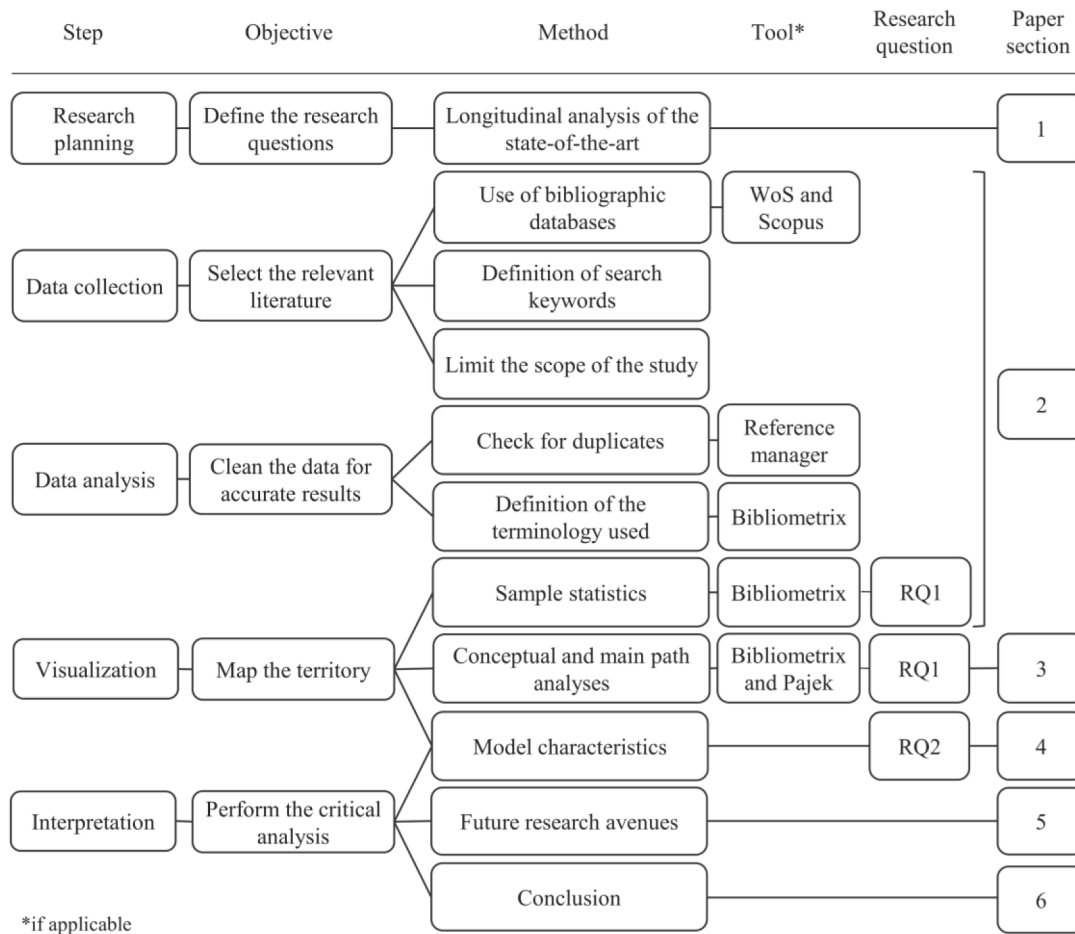


Fig. 1. Science mapping methodology followed by this paper.

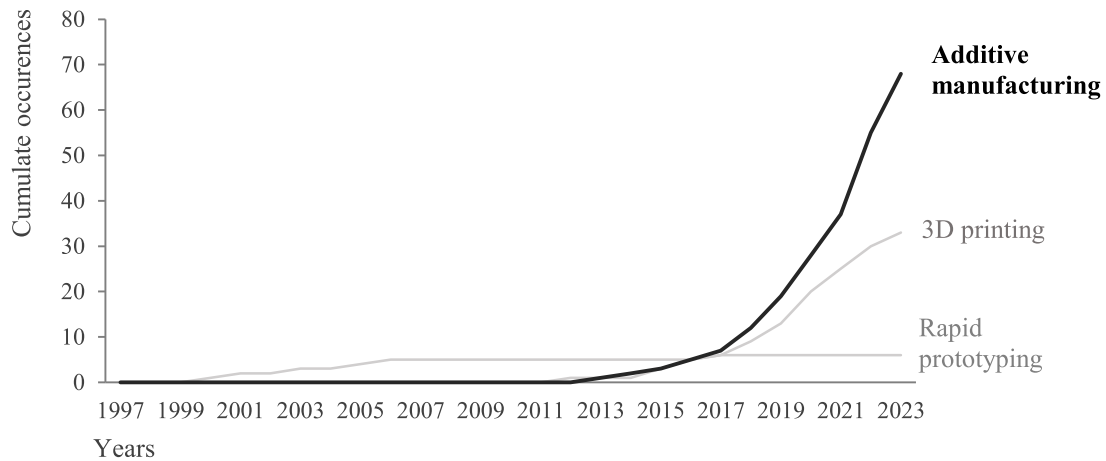


Fig. 2. Terms referring to additive manufacturing over time.

plethora of different terms exists. These terms are listed in Table 1. For example, as mentioned earlier, the knapsack and bin-packing problem exhibit some differences, even though they fall under the terminology of nesting.

2.2. Sample statistics

The annual scientific publication of papers is given in Fig. 3. After the seminal paper of Ikonen et al. (1997), there has been wide neglect for

over a decade. This scenario significantly changed after 2017, highlighting the need for a review to consolidate this field and guide its progression. Furthermore, the increasing adoption of AM machines in real production systems (Fera et al., 2020), has put the need for new models and methodologies capable of effectively handling the related technologies at the forefront of AM-related research.

Fig. 4 summarizes the five principal journals that have published papers on nesting and scheduling problems in AM. As expected, leading industrial engineering, operations research and management, and

Table 1
Terminology applied in this paper.

Terminology	Context-related terms
<i>Additive manufacturing</i>	Direct digital manufacturing; layered manufacturing; rapid prototyping; 3D manufacturing; 3D printing
<i>Nesting</i>	Batch placement; bin-packing; knapsack; grouping parts; layout problem; layout planning; packing; placement
<i>Scheduling</i>	Batch machine scheduling; dispatching; multi-task scheduling; part-to-printer assignment; production planning; sequencing; task-assignment

supply chain management journals feature heavily in this list. Additionally, two of the journals are focused on advanced manufacturing and systems: *International Journal of Advanced Manufacturing Technology* and *Additive Manufacturing*.

3. The nesting and scheduling problems

The rationale of this section is to map the research structure of nesting and scheduling problems in AM, and thus, responding to RQ1. In this study, the analysis extends conventional qualitative reviews, as we leverage the qualitative study through two well-accepted bibliometric methods - conceptual analysis (Aria and Cuccurullo, 2017) and main path analysis (De Nooy et al., 2018). Conceptual analysis is utilized to connect the fundamental concepts of the research field (Zupic and Čater, 2015). Specifically, it can be used to address the different decision types of the nesting and scheduling problem and its related problems. The main path reveals connections among core papers, providing significant details on the relative influence of the publications (De Nooy et al., 2018).

3.1. Conceptual analysis

The conceptual analysis allows for a comprehensive exploration of the actual content of the papers to design the research structure of the field. Here, we focused on the decision types related to both nesting and scheduling problems to arrange the literature. Decision types refer, in this context, to the splitting of the main problem into specific decisions. Through these individual decisions, we gain valuable insights into the subtle complexities of the decision-making process within nesting and scheduling modeling. This directed the analysis towards six decision types, originating four from the nesting and two from the scheduling problem. The decisions are represented in Fig. 5 and associated to the entities: *Part*, *Build* and *Machine*. The entity *Part* is defined as the input of the nesting problem, and the resulting outcome is established by the entity *Build*. In the scheduling problem, the *Build* takes on the role of the input, and the *Machine*, with its schedule, is the output.

3.1.1. Decision types

The first decision type is referred to as *Build formation*. Although there are significant differences in processing techniques between traditional manufacturing processes and AM, it is noteworthy that the batch in BPM problems and the build in AM problems share similarities, albeit with some necessary adaptations. In AM, multiple parts may be grouped based on similar characteristics and produced simultaneously in the same machine (De Antón et al., 2022). The decision of grouping parts that share common characteristics, such as identical heights (Zhang et al., 2020; Yau and Hsu, 2022) or materials, directly impacts the production time and cost (Yau and Hsu, 2022).

In traditional manufacturing, build formation decisions have been usually modeled as one-dimensional knapsack problem (Oh et al., 2020), which means that the batch is reduced to a value, such as maximum weight, area, or volume. In AM, however, the machine

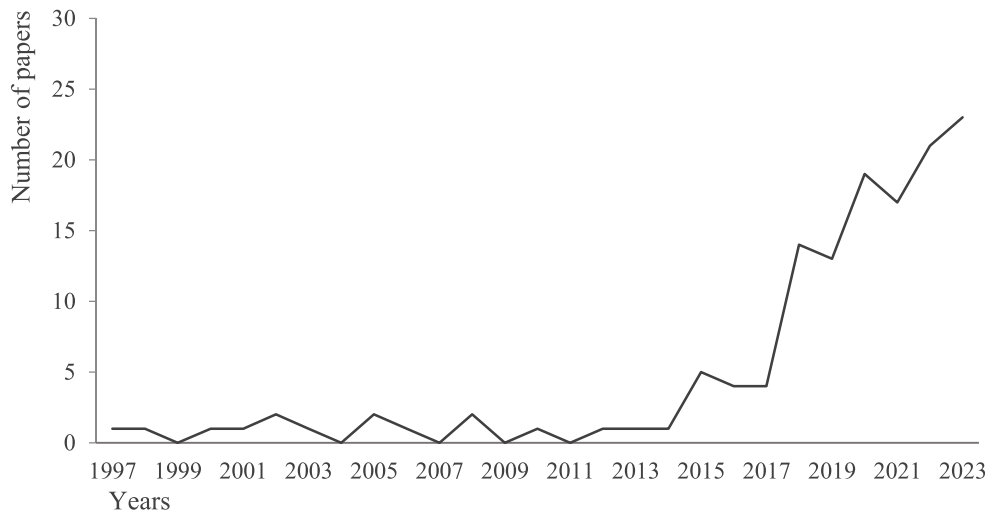


Fig. 3. Annual scientific publication (1997 – December 18, 2023).

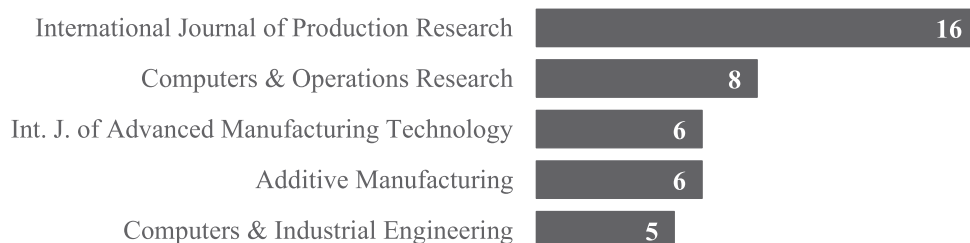


Fig. 4. Top 5 journals publishing papers.

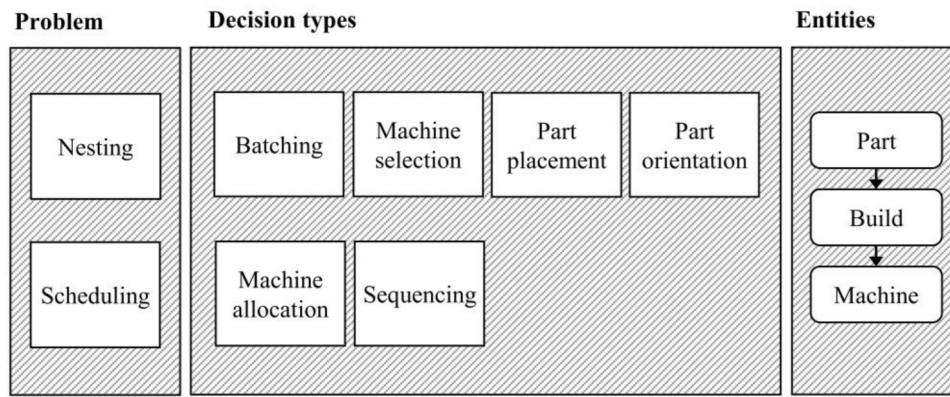


Fig. 5. Decision types of the nesting and scheduling problem in AM.

capacity can be modeled as a 2D or 3D bin-packing problem. FDM and SLA are two technologies that require support structures. These structures are created simultaneously and from the same material as the parts. They serve to prevent part drifting and deformations. If a part is placed over another the support structure of the top part will interfere with the geometry of the top bottom part, at its top surfaces. This interference can significantly degrade the surface and possibly the dimensional accuracy of the bottom part, as well as increase the required post-processing time (Canellidis et al., 2013). For this reason, in FDM and SLA, parts are usually placed alongside each other; hence the bin-packing problem essentially becomes 2D. Other technologies, such as SLS, do not need support structures, and the machine capacity can be modeled as 3D.

The second type of decision is the *Machine selection*. Each build is matched with AM machines whose dimensions are larger than the dimensions of the build. A major concern is the machine performance for producing a specific build (De Antón et al., 2022). Therefore, a set of efficiency criteria can be considered to select machines to builds. For example, one might choose the machine that offers the highest speed among the set of options or opting for the machine that provides the most robust quality assurances (Canellidis et al., 2013). Calabrese et al. (2022) addressed this problem by providing a decision-making framework for choosing feasible machines according to production quality criteria. Maranhã et al. (2023) provide a classification and decision-support framework for selecting AM technologies based on machine characteristics, process features, physical properties of parts, and cost.

In some studies the nesting problem is reduced to these two decision types (Kucukkoc, 2019; Yilmaz, 2020; Griffiths et al., 2019; Dickinson and Knopf, 1998). Authors utilize machine capacity constraints to ensure that the dimensions of the build do not exceed the dimensions of the machine. Nevertheless, to what extent can we claim that they have adequately addressed the nesting problem? All parts must be placed on the build so that they are fully contained within it and do not overlap. Hence, due to the modeling of overlapping needs among parts, issues related to the placement and orientation of the parts have the most substantial influence on the nesting problem (Griffiths et al., 2019).

Part placement responds to where parts can be positioned within the build boundaries, while ensuring there is no overlap between them. Dickinson and Knopf (1998) separate this decision type into parallel and serial placement. In the parallel placement, all parts are placed at once, whereas in the serial placement a rule will determine where the next part will be placed. *Part orientation* is often referred to as one of the most critical decisions within the nesting problem (De Antón et al., 2020). It specifies the best orientation policies that optimize the AM machine utilization, production time, cost, and quality of each build. Hence, each part can be rotated around X-, Y-, and Z-axes to determine the build orientation.

Once all the parts are placed and oriented in the builds, the nesting

problem is concluded, and we move on to the scheduling problem. This problem can be distinguished into two main decision types: *Machine allocation and Sequencing*. In machine allocation, the availability of the machines is checked so that each build can be assigned to one machine (De Antón et al., 2022). One machine can be assigned to several builds (single machine configuration). However, multiple machines in parallel are able to reduce production times and increase AM's competitiveness over the traditional manufacturing processes (Kim and Kim, 2022). Indeed, there are two possibilities of AM parallel configurations: identical parallel and non-identical parallel machines. The latter can be characterized as heterogeneous machines, where each has its specifications, like different cycle times, setup times, and capacity constraints (Che et al., 2021). Some studies have addressed pre- and post-processing operations, which can be incorporated into flow-shop and job-shop settings.

In practice, pre- and post-processing stations can significantly differ from process to process. For example, Power Bed Fusion (PBF) technology require long cool-down times before the parts are removed (Maranhã et al., 2023; Makanda et al., 2022). While in Vat photopolymerization, it is possible to remove the parts almost as soon as it is finished (Makanda et al., 2022). Hence, pre- and post-processing operations can significantly impact the production time. In this context, Kapadia et al. (2022) and Rossi and Lanzetta (2020) tackled flow-shop and job-shop systems, respectively. Aloui and Hadj-Hamou (2021) consider both pre- and post-processing times. Arık (2022) includes downstream post-processing times by considering assembly operations. Thüner et al. (2021) proposed workload control to provide a control solution for the post-processing shop.

Finally, the sequence of builds on each AM machine has to be determined. This is referred to as *Sequencing*. The sequencing plan specifies the order in which builds are to be processed in the machines, so that they meet certain production constraints, including capacity, production levels, precedence, start and due dates, machines capabilities, machine availabilities, build restrictions, machine requirements, and machine availabilities. The sequencing plan has a huge impact on the performance measures of the scheduling problems.

3.1.2. Interdependent problems

Interdependent problems, in this context, refers to adjacent optimization problems that have been integrated with the nesting and scheduling problem. This section aims at extending the analysis beyond the above-mentioned decision types and study a wider spectrum of inter-related problems, that have the potential to affect or be affected by nesting and scheduling problems. For the sake of completeness, the existing literature is arranged into operational and tactical planning levels. By distinguishing the operational and tactical levels, researchers can integrate various insights from different planning functions and timescales. This steered the analysis towards two problems at the

operational level: *Part decomposition* and *Order acceptance*; and one problem from the tactical level: *Part distribution*. Fig. 6 illustrates the interdependent problems found within the literature of nesting and scheduling in AM.

At the operational level, part decomposition widens the research scope to a precedent nesting-related problem. This problem can be used to split a larger part that in its current form exceeds the dimensions of the build space into multiple smaller parts (Oh et al., 2018); it can also be referred to as the printability problem (Yao et al., 2015). Part decomposition is also used to maximize productivity (production time, cost, and part quality); mechanical functionality; artistry; interchangeability; and to avoid the need for support structures (Thürer et al., 2021; Yao et al., 2015).

Several part decomposition methods have been developed in the literature. For example, Luo et al. (2012) suggested a framework to decompose a larger object into smaller dependent parts (sub-assemblies of the original object), known as *Chopper*. In their study, the focus was only on part decomposition, with no consideration of nesting concerns. Yet, some studies have addressed both decomposition and nesting challenges. For example, Chen (2015) proposed a *decompose-and-pack* (*Dapper*) algorithm to decompose an object into minor parts which can be efficiently packed. The Dapper algorithm aims to minimize support material, build time, and the assembly cost. Yao et al. (2015) developed a *partitioning-and-packing* algorithm, in which the partitioning component is based on a level set representation. Demir et al. (2018) introduced a *divide-and-conquer* approach, wherein the parts are derived through a near-convex decomposition. For a detailed and complete understanding of part decomposition, refer to the review of Oh et al. (2018).

Order acceptance can be related to the scheduling problem and occurs whenever the demand exceeds the available capacity of the resources, requiring the rejection of some orders (Kapadia et al., 2022). Hence, determining the set of accepted orders is of great importance within the broader decision problem. Nevertheless, in order acceptance, there exist two distinct scenarios. In the first and most prevalent scenario, the set of orders from different customers is known a priori (Kapadia et al., 2022). In the second scenario, referred to as dynamic, not all orders are known at the beginning of the decision period (Li et al., 2019). For example, Li et al. (2019) consider the order acceptance and scheduling problem with dynamic orders arrival where a part is considered accepted if it can be produced within one of the machines and completed within the delivery time requested by the customer. Kapadia et al. (2022) proposed an order acceptance and scheduling model considering placement and orientation issues, and post-processing requirements. Tafakkori et al. (2022) suggested a framework that considers negotiating with customers to change their desired due dates and assess the influence on the order acceptance and scheduling. For a thorough and comprehensive understanding of order acceptance and scheduling, refer to the taxonomy and review of Slotnick (2011).

Lastly, at the tactical level, scheduling issues have been integrated with the *Part distribution* problem. In this context, a set of orders has to be

manufactured and transported by vehicles to customers (Yilmaz, 2020). While scheduling is vital for achieving efficient production, aligning it with distribution planning enables the minimization of transportation and inventory costs (Li et al., 2022). Solutions to this problem are characterized by production schedules that indicate the specific date, time and manufacturing location for customers' orders as well as the distribution plan that satisfies constraints related to the availability of delivery vehicles, their capacities, and customer due dates (He et al., 2022). Although the integration of production and distribution planning has a considerable impact on supply chain performance, we found that little research attention has been paid to investigate such problems so far. Yilmaz (2020) were among the forerunners in exploring the integration of production and distribution aspects of the AM supply chain. Li et al. (2022) studied the production and distribution problem of spare parts considering two types of customers based on the urgency of delivery times. The authors suggest two different transportation policies to address the delivery time requirements of the two customers. He et al. (2022) integrate the production and distribution problem of spare parts with *Just-In-Time* (JIT) delivery times.

Overall, as nesting and scheduling problems have grown in complexity, additional issues have surfaced, unavoidably expanding the scope of their integration with other decision problems. Still, we note that the extent of integration remains somewhat limited. While the operational level has been extended to encompass two concerns related to nesting and scheduling (part decomposition and order acceptance), the introduction of AM processes in industrial settings poses new challenges for the operational planning.

3.2. Main path analysis: Nesting and scheduling research backbone

This section establishes the connections between the critical papers in an attempt to represent the backbone of the field (De Nooy et al., 2018). The main path analysis provides sound information about the relative influence of specific publications (Zupic and Čater, 2015). It is also examined over time, which is helpful for detecting research traditions and for leveraging prospective analyses (Aria and Cuccurullo, 2017). The combination of conceptual and main path methods provides an appropriate tool to fully understand the research structure of nesting and scheduling problems in the context of AM.

Citations indicate to some degree the scientific relevance of papers as well as their authors and journals. Citation network analysis can consequently be seen as an itinerary carrying scientific knowledge. The main path analysis is a unique method for citation analysis that explicitly focuses on the scientific progress of the field over time and illustrates the evolution of research (De Nooy et al., 2018). For this, each particular citation is given a traversal weight that indicates the extent to which a citation is needed for connecting papers. The search path count method was utilized to quantify the traversal weights and extract the paths. The path with the highest traversal weight is the main path. Fig. 7 presents the main path obtained using the Pajek key-route algorithm (De Nooy et al., 2018), enriched by qualitative analysis of seminal studies in the

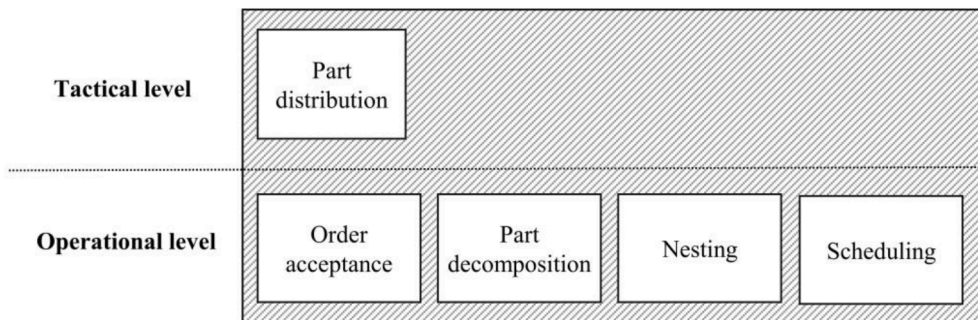


Fig. 6. Interdependent problems with nesting and scheduling in AM.

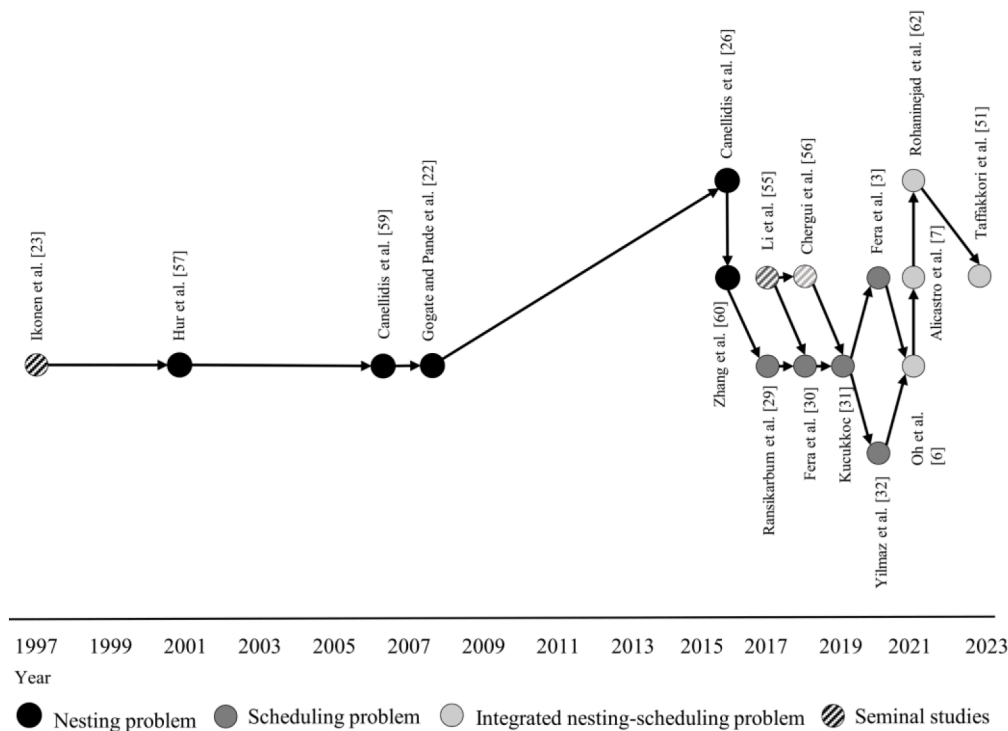


Fig. 7. Main path of nesting and scheduling problems in AM.

field. Indeed, the main path obtained from Pajek shows the most co-cited studies, but this may not be the study from which an idea originated. For example, [Li et al. \(2017\)](#) was the first to define a function in order to calculate a processing time for a batch on a SLM machine, which certainly had an influence on other studies, including [Fera et al. \(2018\)](#), who aimed at minimizing cost on a single SLM machine. Although [Ransikarbum et al. \(2017\)](#) lies on the main path it addressed a different additive manufacturing technique, FDM, where the processing time of the build is known beforehand. The seminal studies are introduced in [section 3.2.1](#) before the main path analysis is further discussed.

The figure represents the backbone of the research on nesting and scheduling problems since its inception. First the literature considered nesting problems. This occurred during a period when AM processes were primarily applied for rapid prototyping, and scheduling needs were not relevant in the context of AM. It was with the improvement of the capabilities of AM that these technologies have become suitable for industrial applications, which in turn augmented the importance of scheduling problems. Nevertheless, scheduling problems are often modeled as one-dimensional knapsack problems ([Oh et al., 2020](#)), and therefore, it was necessary to model the capacity of AM machines through bin-packing problems, which lead to the formation of the integrated nesting-scheduling problem.

3.2.1. Seminal studies

[Ikonen et al. \(1997\)](#) is a seminal paper of the main path. The authors develop a Genetic Algorithm to solve a 3D nesting problem, in which parts are represented as non-convex geometries, including holes and cavities where other smaller parts can be placed. This makes the relations between those parts very complicated. Intuitively, the authors place large parts first and fill the remaining space with smaller ones (this falls under the category of the serial placement problem). A second decision was the orientation of each part in the SLS machine. Each part had 24 allowed orientations (45 degrees of increment around each coordinate axis) with finite relative positions constrained by five attachment points, which define the points where the next part can be placed.

[Li et al. \(2017\)](#) is a seminal paper that introduced the problem of planning and scheduling in AM. The authors considered an objective

function based on the production cost per unit volume of material on non-identical SLM machines. The optimum allocation of parts into such AM machines was solved by a MIP model, and two heuristic procedures - best fit and adapted best fit - are developed. Meanwhile, [Chergui et al. \(2018\)](#) were the first to integrate the planning and scheduling problem in AM with nesting concerns. In this work, a series of order due dates were considered and allocated to identical parallel machines. In contrast to the studies that preceded it, in this model the time spent on the AM machines is expressed in terms of part's total production area and material volume. The objective of the study is to minimize the tardiness and maximize the utilization of the machines. To solve the proposed model, the authors developed a heuristic based on the EDD rule.

3.2.2. Nesting problem

Following the paper of [Ikonen et al. \(1997\)](#), [Hur et al. \(2001\)](#) applied a modified Bottom-Left approach. This is a placement rule that implies positioning parts as close as possible to the bottom left-hand corner of the machine workspace ([Araújo et al., 2020](#)). This approach was implemented by a Genetic Algorithm. Subsequently, the orientation of the parts is optimized by considering their heights to achieve the minimum build height, reducing the production time and improving the surface quality. The authors contend that adopting the minimum build height as the objective function for problems involving dependent parts, resulting from the division of a larger part (part decomposition problem), is the most fitting choice, given the rough surfaces that emerge from the part decomposition process.

While the previous studies of the path have focused on the 3D nesting problem, [Canellidis et al. \(2006\)](#) proposed a 2D problem applied to SLA. In comparison to the SLS process, SLA requires support structures, and therefore, nesting solutions where parts are placed on top of other ones are unfeasible. The use of support structures increases the build time, as this includes the time needed to actually produce the parts and the time required for support removal, and surface finishing. In addition, the authors do not concern the actual geometries of the parts (irregular geometries) but rather minimum bounding boxes, making the problem more manageable and computationally less burdensome. In order to minimize the overall build time, a Left-Border-Down-Border rule was

applied by a Genetic Algorithm for a serial placement solution. Later, the authors extended their method (Canellidis et al., 2013) by approximating the parts geometry through polygons instead of the common bounding box. A No-Fit-Polygon is employed as an additional placing rule based on the former Left-Border-Down-Border.

Gogate and Pande (2008) proposed a 3D nesting problem in which the part orientation problem is optimized based on the user criteria. This is operationalized by giving arbitrarily weights to each of the evaluation criteria in the objective function, including build height, average staircase error, volume of support structures, and contact interface of parts and support structures. By assigning different weights to interpolate between the different evaluation criteria, the problem can be customized to meet the functional requirements of several AM machines. Accordingly, the Bottom-Left method is implemented by a Genetic Algorithm. Parts are represented through voxel models and could be rotated around the build axis with an increment of 45 degrees.

Zhang et al. (2016) addressed the 2D problem to solve the nesting problem through a two-step algorithm: first a AM feature-based orientation involves determining the optimal build orientation for each part to ensure high production quality, and second a parallel optimization model is used to maximize the compactness of the nesting problem through the use of polygons to represent the geometry of the parts. To solve this problem, a Genetic Algorithm was used, which reinforces the idea that this metaheuristic is a competitive solution technique for solving nesting problems.

Nesting problems were the first AM decisions tackled by researchers, which is partly explained by the primary need to build multiple prototype parts simultaneously on a single building platform (Hur et al., 2001). As AM matured, there was an increasing need to be able to not only manufacture prototypes to facilitate rapid product development but also directly manufacture functional or end-use parts for specific applications (Zhang and Bernard, 2014). Owing to its advantage of manufacturing complex parts (surpassing traditional processes), AM has played a significant role in customized production. For all these reasons, nesting problems are critical to the AM planning decision-making.

3.2.3. Scheduling problem

In comparison to the nesting problem, the focus here is not on achieving the optimal placement and orientation of multiple parts on a build but rather on finding the best allocation and sequencing of builds in the AM machines (Oh et al., 2020). Li et al. (2017) paved the way for this problem (see section 3.2.1), leading to the subsequent studies indicated in our main path. For example, Ransikarbum et al. (2017), who consider a set of non-identical parallel FDM machines, and used a Mixed-Integer Linear Programming (MILP) model to assign a build per machine with the objectives of minimizing cost, maximizing load balance among machines, minimizing the total tardiness and minimizing the total number of unprinted builds.

Fera et al. (2018) addressed the single-machine scheduling; a set of builds were sequenced in a SLM machine, and a multi-objective model was formulated to minimize the production cost and the weight of the total earliness/tardiness. In their approach, the schedule that meets due dates with the least cost is solved using a modified Genetic Algorithm.

Meanwhile, Kucukkoc (2019) provides a stepping-stone to model advances for AM scheduling, since the authors addressed three different SLM configurations: single machine, parallel identical machines, and non-identical parallel machines. The different configurations were tackled through three MILP models to minimize the makespan, and the optimum solutions have been reported. The results showed limited applicability when problem size increases, which indicates the need for effective heuristics and metaheuristics to solve large problem instances.

The main path analysis also reveals that, at some point, the literature on nesting and scheduling is branched into two research streams. One stream is dominated by Fera et al. (2020), which represents the most evident research trajectory when compared to the previous papers. The other stream is led by the paper of Yilmaz (2020), which explores the

scheduling together with other supply chain functions. Later on, Oh et al. (2020) integrated both research streams in a comprehensive review on nesting and scheduling problems in AM.

Fera et al. (2020) developed an improved Tabu Search to solve the same problem proposed in Fera et al. (2018). They showed that their algorithm outperforms the earlier modified Genetic Algorithm in solving small to large size instances. Yilmaz (2020) took the pioneering approach of investigating the combination of production and distribution functions in an AM supply chain to propose a two-stage scheduling problem, in which parts are produced on SLM machines by one manufacturer in the production step and then distributed to various customers by means of solving a transportation problem. A heuristic procedure with five selection rules was proposed to minimize the makespan, and computational experiments were conducted using problem instances generated for small and large size problems.

The need for solving AM scheduling problems results from an increasing adoption of AM machines in real manufacturing contexts (Fera et al., 2020). This shift asked for innovative models and methods capable of efficiently dealing with AM technologies. While the nesting problem looked at the AM problem in the realm of a single machine, this second problem comes as a step forward in bringing the AM planning closer to practical scenarios. Nevertheless, the models that address this problem are often modeled as one-dimensional knapsack, and thus it was necessary to model the capacity of AM machines through nesting issues, which lead to the development of the integrated nesting-scheduling problem.

3.2.4. Integrated nesting-scheduling problem

Although the integration of nesting and scheduling problems has been only recently addressed, it remains of utmost importance to model the interdependence between these decisions. Following the research by Chergui et al. (2018), which is a seminal paper for integrated nesting and scheduling (see section 3.2.1), the comprehensive review of Oh et al. (2020) holds a pivotal position in our main path because it was the first review paper that has thoroughly combined both nesting and scheduling papers in a holistic manner. The authors proposed a taxonomy to classify nesting and scheduling problems into four classes depending on *Part*, *Build*, and *AM Machine*. In the M/S/S class, problems consider multiple parts grouped into a single build and assigned to a single machine. M/M/S class covers problems where a single AM machine sequentially processes multiple builds, whereas in the M/M/M class a set of identical or non-identical machines are assigned to builds. Finally, if the optimization problem considers that only one part is assigned to each machine for a build cycle, these are classified into -/M/M class. Hence, by comprehensively covering nesting and scheduling problems in AM, the developed taxonomy complements the literature with a hierarchical perspective.

Faced with the hardness of solving both nesting and scheduling problems, Alicastro et al. (2021) used a Q-learning variable neighborhood search to develop a Reinforcement Learning Iterated Local Search metaheuristic. The nesting problem discussed in their paper is called a strip-packing problem, which can be described through a rectangular bin with finite width and infinite height, in which the objective of this problem is to minimize the nesting height. In pursuit of simplicity, the authors considered bounding boxes to represent the parts. Thus, the problem is solved through a level-oriented method where parts are arranged in rows forming levels, such that in each level the set of parts are bottom-aligned. For the scheduling problem, non-identical SLM machines are considered to minimize the makespan.

Rohaninejad et al. (2022) addressed a non-identical SLM machines environment. Differing from the earlier efforts, this study did not investigate single-material part production, but rather SLM machines capable of working with different powder types, including steel, iron-, aluminum-, and titanium-based alloys. Hence, the parts belonging to a particular job must be of the same material type, while different material types can be used for successive builds. To produce a sequence of builds

with different material types, it is necessary to clean the SLM machine as well as to replace the related powder. This setup process requires additional time, and the setup time is sequence-dependent. Accordingly, a bi-objective MILP model was proposed considering the makespan and the total tardiness cost, and a hybrid metaheuristic was developed combining the Non-Dominated Sorting Genetic Algorithm (NSGA-II) with a learning-based local search founded on a k-means clustering algorithm and a regression neural network.

While previous papers have been broadening the efforts to encompass mainly economic goals, Tafakkori et al. (2022) have been at the forefront of the research addressing sustainability concerns in AM. In their study, a multi-objective model was developed on the basis of the triple bottom line that includes profit, energy utilization of machines, and goodwill losses as the three sustainability criteria. Beyond part placement, the part orientation problem along with irregular geometries were considered into the nesting problem, and a non-identical parallel machine scheduling was solved. An order acceptance problem, in which the manufacturer has set some negotiation strategy to win orders based on increasing due date desired by customers for price reductions, were also considered. The model was solved by three algorithms: robust improved ϵ -constraint method, NSGA-II, and the multi-objective metaheuristic Grey Wolf Optimizer.

3.3. Critical remarks on the main path analysis

Although the main path analysis of nesting and scheduling in AM

provides a thorough view about the influence of each specific publication (Zupic and Čater, 2015), it is crucial to recognize that the AM field's relative youth and rapid pace of progress may not be adequately captured in the citation network. To address this shortcoming, we included in the main path a secondary path analysis. The secondary paths point to pertinent research that ultimately contributes to draft a complete research territory. To conduct this analysis, we examined each pair of interconnected papers to uncover secondary streams gaining relevance within the nesting and scheduling domain. Fig. 8 shows two other paths - processing operations and collaborative manufacturing - alongside the main path presented above.

3.3.1. Collaborative manufacturing

This research stream introduces the collaborative multi-site production problem in AM, wherein orders must be batched and scheduled at several geographically dispersed manufacturing sites. Given the unique characteristics of AM, this problem is particularly relevant. One of the strengths of AM is that apart from being a toolless production technologies, the entire production process is handled by a machine-readable format, i.e., each phase of the manufacturing process is digitally captured and can be readily interpreted by machines or computer systems. These advantages allow the digital transfer of production batches from one machine to another, without the need for costly transfer methods (Zehetner and Gansterer, 2022). Therefore, firms could easily exchange orders among manufacturing plants to balance the use of facilities or even save transportation costs, as one manufacturing site

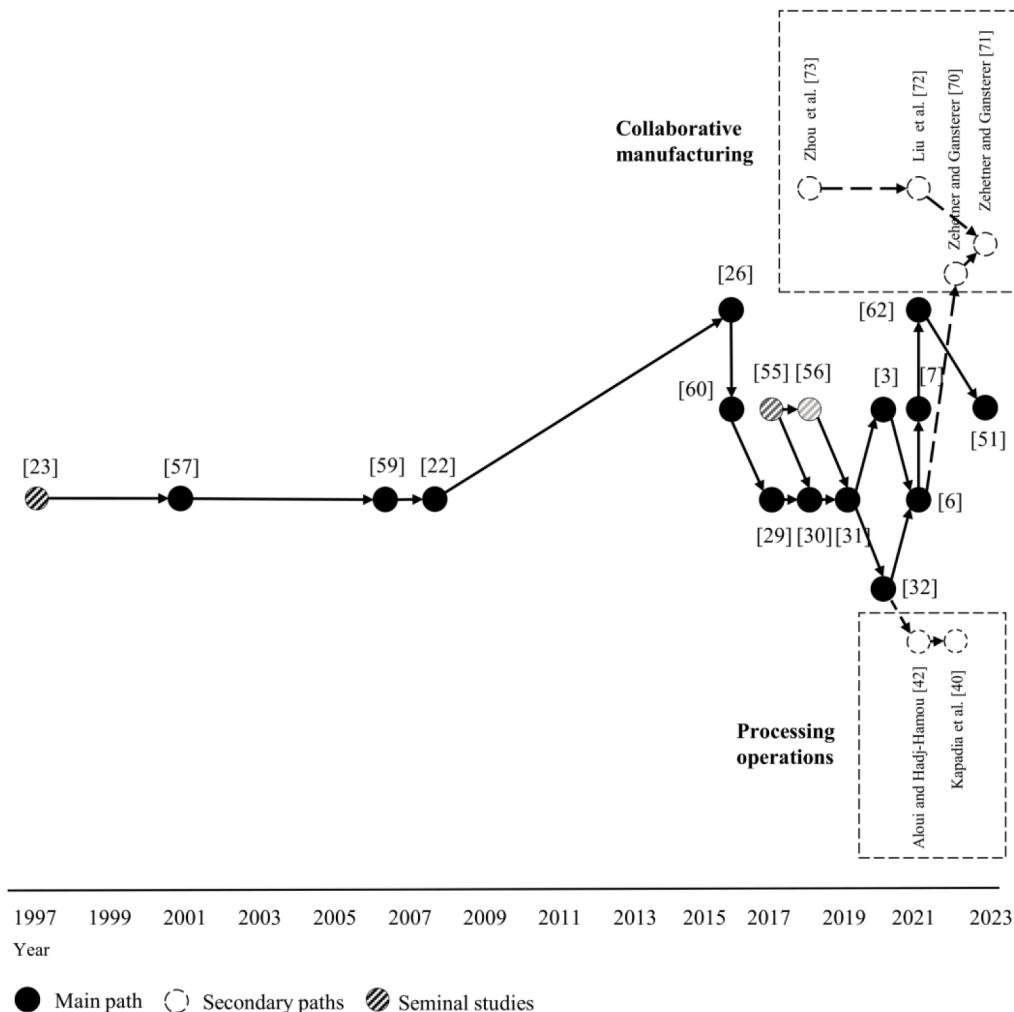


Fig. 8. Secondary paths of the nesting and scheduling problem in AM.

may be situated closer to customers.

Zehetner and Gansterer (2022) were the first to introduce the collaborative multi-site production problem in AM. They developed a two-step approach, wherein the overall problem is decomposed into a nesting and a scheduling problem. Two different Genetic Algorithms were proposed for the nesting problem and a MILP model for the scheduling problem. Their following work (Zehetner and Gansterer, 2023) studies the relationship between collaborative production and Cloud Manufacturing (CMfg), since CMfg is considered an essential enabler for achieving collaborative production goals. The research conducted by Zehetner and Gansterer (2023) was influenced by the work of Zhou et al. (2018) and Liu et al. (2021). Zhou et al. (2018) focused their study on the supply-demand matching problem for AM services in CMfg. Different service attributes were considered, including model size, material, accuracy, cost, time, and transportation. To minimize the delivery time from service providers to customers, a Genetic Algorithm was proposed. Likewise, Liu et al. (2021) developed a non-cooperative game model of AM services in a CMfg. The objective is to reduce the makespan, cost and to maximize the service quality.

CMfg can be seen as a technology-driven approach that uses cloud computing and networking technologies to facilitate the sharing of manufacturing resources, information, and capabilities over a network (Zhou et al., 2018). It acts as a technological infrastructure that supports efficient management and operation of the manufacturing process. AM can serve as a catalyst and demonstration of CMfg applications (Liu et al., 2021). At the same time, CMfg is a key technology to leverage AM systems, connecting suppliers to customers and to the CMfg operator, who is responsible for managing and operating the CMfg platform (Wang et al., 2019). Although collaborative manufacturing is a complex decision problem, for the convenience of AM research, the scheduling problem related to CMfg has been studied previously (Liu et al., 2021). Hence, the introduction of AM within CMfg is a sound direction to further improve the utilization of idle resources and achieve cost reductions (Wu et al., 2022).

Recently, there has also been a proliferation of research on collaborative multi-robot manufacturing (Jin et al., 2019; Poudel et al., 2020; McPherson and Zhou, 2018; Poudel et al., 2021; Shen et al., 2019; Jiang et al., 2021; Weber et al., 2023). These studies suggest a new approach to provide scalability to AM, allowing multiple robots to cooperate on a single printing job (Shen et al., 2019). A large part is first divided into multiple chunks (divisions), which are then assigned to multiple robots to work cooperatively (Poudel et al., 2021). In this scenario, typical decisions include geometric partition of the large part into chunks, scheduling of robots to chunks, path planning (collision-free moving path planning for robots to transition from one workstation to another) and motion planning (collision-free infill paths generation for neighboring robots to work together during manufacturing).

3.3.2. Processing operations

Pre- and post-processing operations have a substantial impact on the AM scheduling problem, namely when the parts have due dates and fixed delivery times. Aloui and Hadj-Hamou (2021) introduced two models for Power-based Laser and Multi Jet Fusion (MJF) technologies to estimate the production time, including file and machine preparation, layer construction, and post-processing operations (finishing, sorting and packaging). The objective is to reduce the total delay of the parts to be produced and to maximize the use rate of the AM machines. The authors solved the problem with an exact method using Mixed-Linear Programming and a heuristic procedure based on the Earliest Due Date (EDD) rule and the shortest estimated manufacturing time.

Meanwhile, Kapadia et al. (2022) considered the problem of order acceptance and scheduling, satisfying all technological constraints, including part placement and part orientation into the AM machine and post-processing operations for support removal and surface finishing. The authors used a Genetic Algorithm to select orders for complete or partial acceptance and to produce machine schedules such that the

profit is maximized. Specifically, post-processing operations pose significant challenges for production scheduling since routings and processing requirements vary, and therefore parts produced at AM machines often require alternative post-processing operations and in different sequences. Thüerer et al. (2021) considered the use of workload control in a flow-shop where nine distinct pool sequencing rules aimed at minimizing shop floor throughput times and preventing starvation. Once a job is released and enters the queue at the AM station, three dispatching rules prioritize jobs awaiting processing in the queue of the station.

The work carried out in this stream suggests that explicit consideration of pre- and post-processing operations can significantly improve the performance of AM scheduling, while also approaching practical needs. Particularly, the delivery time seems to be largely determined by post-processing stations, which may significantly constrain the effectiveness of AM (Thüerer et al., 2021). These stations, cause difficulties for production planning and, as a result, problems in adhering to promised due dates.

4. Model characteristics

To answer our second research question, we review the main characteristics of the mathematical formulations for nesting and scheduling problems in AM. Model variants can be attributed to the different levels of detail and industrial contexts under consideration. Our analysis was centered on seven fundamental model aspects. *Part placement* and *part orientation* decisions were brought to the forefront of the discussion, since both have a great level of influence on the nesting problem. Given the different *machine* configurations addressed in the literature, we adopt a categorization considering single-machine, identical parallel machines, non-identical parallel machines, flow-shop, and job-shop. Machine allocation and sequencing decisions were considered main features of the *scheduling* problem. Finally, optimization issues are detailed through the *assumptions* or *constraints*, the *objective* function, and the *solution approach* used in the models. Table 2 shows the various assumptions used in the literature. The models were categorized based on the problems of nesting (N), scheduling (S), part decomposition (D), part distribution (T), and order acceptance (OA). Table 3 classifies the studies through the above-mentioned aspects.

To facilitate comprehension of Table 3, “✓” indicates that the aspect was covered in the paper, while “×” denotes that the aspect was not addressed in the paper. In addition, to properly interpret the column of

Table 2
Assumptions of nesting and scheduling problems in AM.

Code	Assumptions or constraints
N1	The orientation of parts is pre-determined
N2	Parts are not allowed to stack on top of each other (2D nesting)
N3	Parts are allowed to stack on top of each other (3D nesting)
N4	Parts are placed at once (parallel placement)
N5	A rule determines where the next part will be placed next (serial placement)
N6	Parts are approximated by regular geometries (e.g., bounding boxes)
D1	A larger part can be divided into smaller parts to fit in the build (dependent parts)
S1	Sequence-independent setup time (setup time is included in the production time)
S2	Sequence-dependent setup time
S3	The machines are compatible for producing parts with any type of material
S4	Sequence restrictions (e.g., due dates or dependencies)
S5	Uncertainty factors (e.g., arrival times, production times, and machine availability)
S6	The AM machines belong to multiple suppliers in different geographical locations
S7	Hybrid manufacturing systems, combining AM with traditional manufacturing
T1	Each build is transported from the manufacturer to the customers by a vehicle
OA1	The decision-maker can accept or reject orders (order acceptance problem)

Table 3

Framework on the characteristics of the models.

Paper	Placement	Orientation	Machine	Scheduling	Assumptions	Objective	Approach
Ikonen et al. (1997)	✓	✓	Single	×	N3;N4	Part distance	GA
Dickinson and Knopf (1998)	✓	✓	Single	×	N3;N5	Compactness	SA
Hur et al. (2000)	✓	×	Single	×	N1;N3;N5;N6	Compactness	GA
Hur et al. (2001)	✓	✓	Single	×	N3;N5;D1	Compactness	GA
Zhang et al. (2002)	✓	✓	Single	×	N3;N5;N6	Build height	SA
Dickinson and Knopf (2002)	✓	✓	Single	×	N3;N5	Compactness	SA
Canellidis et al. (2005)	✓	×	Single	×	N2;N5;N6	Compactness	SA
Canellidis et al. (2006)	✓	✓	Single	×	N2;N5;N6	Compactness	GA
Gogate and Pande (2008)	✓	×	Single	×	N1;N3;N5	Multi-objective	GA
Yang et al. (2008)	✓	✓	Single	×	N2;N3;N5;N6	Compactness	GA
Kim et al. (2010)	×	×	Single	✓	S1;S3	Material switches	Heuristic
Canellidis et al. (2013)	✓	×	Single	×	N1;N2;N5	Compactness	GA
Wu et al. (2014)	✓	✓	Single	×	N2;N5	Multi-objective	GA
Feng and Jia (2015)	×	×	RM ¹	✓	S1;S4;S7	Multi-objective	DP ³
Freens et al. (2015)	×	×	RM	✓	S4	Tardiness cost	ILP ⁴
Cao et al. (2015)	✓	✓	Single	×	N2;N6	Compactness	SA
Edelkamp and Wichern (2015)	✓	✓	Single	×	N3;N4	Centrality	SA
Zhang et al. (2016)	✓	✓	Single	×	N2;N3	Overlap area	GA
Koblasa et al. (2016)	✓	✓	Single	×	N3;N5;N6	Compactness	Heuristic
Canellidis et al. (2016)	✓	✓	Single	×	N2;N5	Compactness	GA
Li et al. (2017)	×	×	RM	✓	N1;N2;N6	Build cost	MIP;Heuristic
Ransikarbum et al. (2017)	×	×	RM	✓	S4	Multi-objective	MILP
Kim et al. (2017)	×	×	PM ³	✓	D1;S1	Makespan	GA
Zhou et al. (2018)	×	×	RM	✓	S4;S6	Total tardiness	GA
Li and Zhang (2018)	✓	✓	Single	✓	N2;N5;N6	Makespan	MIP ⁵ ;Heuristic
Chergui et al. (2018)	✓	×	PM	✓	N2;N6;S4	Total tardiness	Heuristic
Li and Zhang (2018)	✓	✓	RM	✓	N2;N5;N6;S1	Makespan	Heuristic
Dvorak et al. (2018)	✓	✓	Job-shop	✓	N2;S4	Tardiness	Local search
Oh et al. (2018)	✓	✓	Flow-shop	✓	N2;N5;N6;D1	Total time	GA
Romanova et al. (2018)	✓	✓	Single	×	N3;N5	Part distance	NLP ⁶
Zhang et al. (2018)	✓	✓	Single	×	N2;N4	Overlap area	Heuristic
Gopsill and Hicks (2018)	✓	✓	PM	✓	N2;N6;S3	Productivity	GA;Heuristic
Fera et al. (2018)	×	×	Single	✓	S4	Tardiness & cost	GA
Romanova et al. (2019)	✓	✓	Single	×	N3;N5	Part distance	MIP
Griffiths et al. (2019)	✓	✓	Single	×	N2;N5	Total build cost	Tabu Search
Kucukkoc (2019)	×	×	Single	✓	S1	Makespan	MILP
Kucukkoc (2019)	×	×	PM	✓	S1	Makespan	MILP
Kucukkoc (2019)	×	×	RM	✓	S1	Makespan	MILP
Li et al. (2019)	×	×	RM	✓	S6;OA1	Profit	Heuristic
Litvinchev et al. (2019)	✓	✓	Single	×	N3;N5	Build height	MIP
Chen (2019)	×	×	RM ²	✓	S5;S6;T1	Makespan	FMILP;FMIQP ⁷
Fera et al. (2020)	×	×	Single	✓	S4	Build time & cost	Tabu Search
Chekanin (2020)	✓	✓	Single	×	N2;N5	Part distance	Heuristic
Darwish et al. (2020)	✓	×	RM	✓	N2;S4;OA1	Workload	Heuristic
Valizadeh et al. (2020)	×	×	Flow-shop	✓	S7	Makespan	PSO ⁸
Che et al. (2021)	✓	✓	RM	✓	N2;N6;S1	Makespan	MILP;SA
Li et al. (2019)	✓	×	RM	✓	N1;N2;S5;OA1	Profit	Heuristic
Arndt et al. (2020)	✓	✓	Single	×	N3;N4	Multi-objective	GA
Tavakkoli-Moghaddam et al. (2020)	×	×	RM	✓	S2;S4	Makespan	MILP
Zhang et al. (2020)	✓	✓	PM	✓	N2;N5;S7	Makespan	GA
Ransikarbum et al. (2020)	✓	×	RM	✓	N2;N3;S;S4	Multi-objective	MILP
Yılmaz (2020)	×	×	RM	✓	S1;T1	Makespan	Heuristic
Rossi and Lanzetta (2020)	×	×	Job-shop	✓	S1;S7	Makespan	ACO ⁹
Chugay et al. (2020)	✓	✓	Single	×	N3;N4	Build height	NLP
Romanova (2021)	✓	✓	Single	×	N2;N4	Compactness	MIP
Kim and Kim (2021)	×	×	PM	✓	S2	Makespan	MIP;GA
Chekanin and Chekanin (2021)	✓	✓	Single	×	N3;N5	Compactness	Heuristic
Kucukkoc et al. (2021)	✓	×	RM	✓	N1;N2;N6;S4	Total tardiness	GA
Aloui and Hadj-Hamou (2021)	✓	×	RM	✓	N2;N6;S4;S7	Total tardiness	MIP;Heuristic
Chugay and Zhuravka (2021)	✓	✓	Single	×	N3;N4	Build height	NLP
Rohaninejad et al. (2021)	×	×	RM	✓	S1;S3;S4	Total tardiness	Matheuristic ¹⁰
Darwish et al. (2021)	×	×	RM	✓	S4;OA1	Workload	Heuristic
Alicastro et al. (2021)	✓	×	RM	✓	N1;N2;N6;S1	Makespan	Hybrid ¹¹
Kapadia et al. (2021)	✓	×	Flow-shop	✓	N1;N2;N6;OA1	Profit	GA
Pantoja-Benavides and Álvarez-Martínez (2022)	✓	✓	Single	×	N3;N4	Compactness	Hybrid ¹²
Yang et al. (2022)	✓	✓	Single	×	N2;N4	Compactness	Heuristic
Kim and Kim (2021)	×	×	PM	✓	D1;S3	Makespan	MILP
Garashchenko et al. (2022)	✓	✓	Single	×	N2;N4	Build height	GA
Arik (2022)	✓	✓	Flow-shop	✓	N2;N6;S1;S7	Makespan	MIP;Heuristic
Yu et al. (2022)	✓	✓	RM	✓	N2;N6;S1;S4	Total tardiness	MILP
Toksari and Toga (2022)	×	×	Single	✓	S2;S3	Makespan	MILP
Li et al. (2022)	×	×	Single	✓	S4;T1	Multi-objective	MIP
Makanda et al. (2022)	✓	✓	RM	✓	N2;S6;OA1	Cost and time	Heuristic
Tafakkori et al. (2022)	✓	✓	RM	✓	N2;N6;S4;OA1	Multi-objective	Three methods ¹³

(continued on next page)

Table 3 (continued)

Paper	Placement	Orientation	Machine	Scheduling	Assumptions	Objective	Approach
Wu et al. (2022)	✓	×	RM	✓	N1;N6;S6	Average cost	MILP;Heuristic
Hu et al. (2022)	✓	×	RM	✓	N2;N6;S1	Makespan	MILP;Heuristic
Altekin and Bukchin (2022)	✓	×	RM	✓	N1;N2;N6;S1	Makespan	MILP;Heuristic
Rohaninejad et al. (2022)	✓	×	RM	✓	N1;N2;N6;S4	Total Tardiness	NSGA-II
He et al. (2022)	×	×	Single	✓	T1	Makespan	MIP
Ying et al. (2022)	✓	×	Single	✓	N1;N2;N6;S4	Tardiness	Heuristic
Oh and Cho (2022)	×	×	Flow-shop	✓	D1;S1;S7	Makespan	MIP
Kim and Kim (2022)	×	×	RM ³	✓	S2	Makespan	Four methods ¹⁴
Yau and Hsu (2022)	✓	✓	Single	×	N3;N6;D1	Build height	FPA ¹⁵
Zipfel et al. (2023)	×	×	RM	×	S2;S4	Total tardiness	MIP;ILS ¹⁶
Yang et al. (2023)	✓	✓	Single	✓	N3;N4	Compactness	Heuristic
Lee and Kim (2023)	✓	✓	RM	✓	N2;N5;N6	Makespan	MILP;GA;PSO
Uzunoglu et al. (2023)	×	×	RM	✓	S1;S4	Total tardiness	MILP;Hybrid ¹⁷
Yu et al. (2023)	✓	×	Single	✓	N1;N2;N6	Makespan	MIP
Calabrese et al. (2022)	✓	✓	Single	×	N2;N6	Compactness	Heuristic
Nascimento et al. (2023)	✓	×	RM	✓	N1;N2;S4	Cost; Tardiness	CP ¹⁸
Ying and Lin (2023)	✓	×	Job-shop	✓	N1;N2;N6;S7	Makespan	MILP;Hybrid ¹⁹
Zehetner and Gansterer (2023)	✓	×	RM	✓	N2;N6;S1;T1	Total cost	Heuristic
Lu et al. (2023)	✓	✓	RM	✓	N2;N5;S1	Makespan	Heuristic
Chandra Sugianto and Soo Kim (2023)	×	×	RM	✓	S2;T1	Makespan	PSO
Hu et al. (2023)	✓	✓	RM	✓	N2;N6;S4	Total tardiness	MILP;Heuristic
Lin et al. (2023)	✓	✓	RM	✓	N2;N6	Multi-objective	MILP

¹Non-identical parallel machines; ²Identical parallel machines; ³Dynamic Programming; ⁴Mixed-Integer Programming; ⁵Integer-Linear Programming; ⁶Non-Linear Programming; ⁷Fuzzy Mixed-Integer Linear Programming Model and Fuzzy Mixed-Integer Quadratic Programming Model; ⁸Particle Swarm Optimization; ⁹Ant Colony Optimization; ¹⁰Genetic Algorithm and MIP; ¹¹Reinforcement Learning Iterated Local Search metaheuristic; ¹²Simulation and Tabu Search; ¹³Robust improved ϵ -constraint method; NSGA-II; and Grey Wolf Optimizer; ¹⁴Mixed-Integer Programming; ⁸Particle Swarm Optimization; Genetic Algorithm; and Imperialist Competitive Algorithm; ¹⁵Flower Pollination Algorithm; ¹⁶Iterad Local Search; ¹⁷Learning-augmented heuristic; ¹⁸Constraint Programming; ¹⁹Reinforcement Learning Iterated Greedy Algorithm.

assumptions, it should be cross-referenced with Table 2.

The nesting problem (led by placement and orientation decisions) has been studied with the primary objective of maximizing the compactness of the build. The outcome is evident; by optimizing the compactness more parts within the build are accommodated and more parts are produced during one cycle time (Kucukkoc, 2021). Yet, the pressing question that emerges when considering compactness is how to model overlapping requirements between parts. While some authors model this issue by geometric approximations to represent the real form of the parts, thereby introducing a tolerance between the parts (Fera et al., 2018; Demir et al., 2018; Zehetner and Gansterer, 2022), other authors consider overlap concerns in the objective function (Che et al., 2021; Wu et al., 2022).

Even considering simplifications, such as geometric approximations, the optimization problem is not trivial, and the computational burden increases sharply with the irregularity of the geometries, and hence the model solution time also increases (Araújo et al., 2020). To overcome this situation, it is common practice to simplify the parts to regular geometries, including bounding boxes (Oh et al., 2018), cylinders (Zhang et al., 2002), and spheres (Dickinson and Knopf, 2002). Simplifications, while theoretically useful to reduce the computational burden, are far from practical for real-world scenarios. This has been noticed by scholars who have effectively solved the nesting problem involving irregular geometries. For a thorough and comprehensive understanding of nesting problems with irregular geometries, refer to the noteworthy review of Araújo et al. (2019).

To a lesser extent, the second most often discussed objective, the build height, directly impacts the production time and cost (Yau and Hsu, 2022). This occurs since AM machines operate layer-by-layer, and as the number of layers increases, the build time also grows. Given this premise, some authors consider that the build time corresponds to the production time of the highest part within the build (Edelkamp and Wichern, 2015). Other nesting objectives were considered on an ad hoc basis, including the minimization of the part's distance within the build and the distance between parts and the build boundaries (Ikonen and Biles, 1997).

Regarding placement decisions, two alternative strategies have been discussed in the literature: parallel placement and serial placement. In

the parallel placement, all parts are positioned at once, and therefore, it is evident that parallel placements are usually solved by metaheuristics, including Simulated Annealing or Genetic Algorithms. Otherwise, serial placement decisions are essentially rule-based approaches, making them computationally less demanding, which often leads to their resolution using heuristics.

Ensuring the proper orientation of parts within the build is another critical aspect of modeling. According to the literature, there are two different strategies: two-step and integrated. In the two-step approach, orientations are determined first and then the placement of parts is decided already with the orientations fixed from the preceding phase (Cao et al., 2015). In some models, the first step considers rotations on X- and Y-axes for build orientation, and then in the following step, rotations on the Z-axis (Gogate and Pande, 2008; Che et al., 2021; Li et al., 2019; Aloui and Hadj-Hamou, 2021). In general, the two-step approach provides lower computational complexity than the integrated approach. In the integrated method, the above-mentioned first and second steps are iteratively or simultaneously used to reach an optimized solution. It should be noted that modeling the orientation of parts elevates the nesting problem to the category of NP-hard (Li and Zhang, 2018), which leads some authors to adopt a pre-determined part orientation, and therefore reducing the orientation aspect to a single rotation along an axis (Shen et al., 2019; Jiang et al., 2021; Garashchenko et al., 2022).

In scheduling problems, the primary objective is the minimization of makespan, followed by the minimization of tardiness. In this sense, non-identical parallel machines were well-addressed by the literature, followed by identical parallel machines and then single-machine configurations. Slightly less considered were configurations with upstream and downstream operations (Li et al., 2022; Canellidis et al., 2006). This means that pre-processing and post-processing operations, which are integral stages for the majority of AM systems (Thürer et al., 2021), have been widely neglected when developing scheduling models. It is a novel research stream identified by our main path analysis.

In multi-material AM scenarios, where various materials may need to be used in a single build or production cycle, including setup times becomes essential to address practical scenarios. When solving the scheduling problem, some authors considered sequence-dependent setup times (Rohaninejad et al., 2021). However, this requirement

needs to be further studied in AM.

Another modeling characteristic is the allocation of orders from different customer in an CMfg environment. This scenario is gaining popularity in the AM scheduling problem. Likewise, hybrid manufacturing appears to be a likely scenario in the foreseeable future given the coexistence of AM and traditional processes. However, only a restricted number of studies model the scheduling problem based on the conditions of coexistence, such as Flow-shop and Job-shop configurations.

In terms of used approaches, nesting problems are often addressed using metaheuristic algorithms, being the Genetic Algorithm the predominant technique, followed by Simulated Annealing. Rule-based heuristics were frequently incorporated in these metaheuristics, for example, both Bottom-Left (Hur et al., 2001) and Left Border-Down Border algorithms (Canellidis et al., 2013) are widely used for serial part placement. Exact algorithms were considered to a lesser extent in the domain of nesting.

In the context of scheduling problems, heuristic and metaheuristics methods continue to stand out as the most often used algorithms. Yet, there is an increased application of exact algorithms, including Mixed-Integer Programming (Griffiths et al., 2019; Jiang et al., 2021; Weber et al., 2023) and MILP (Rohaninejad et al., 2022; Hur et al., 2000; Zhang et al., 2002) models. It is important to acknowledge that exact algorithms still have limited practicality when dealing with large-scale optimization problems, like those encountered in industrial-sized scheduling challenges.

Finally, integrated nesting-scheduling models were usually solved by metaheuristic algorithms (Arik, 2022; Zehetner and Gansterer, 2023; Dickinson and Knopf, 2002) due to their complexity. Nevertheless, in recent developments, hybrid algorithms that blend metaheuristics with machine learning techniques have emerged as prominent directions of research to solve the integrated nesting-scheduling problem.

5. Future research avenues

The nesting and scheduling problem has progressed significantly. But there remain relevant issues in this field that still require further attention. The most pressing research avenues to further advance the field are outlined in the following sub sections.

5.1. Closing the distance between modeling and reality

Assumptions specify the conditions under which a mathematical model is expected to be accurate and relevant for decision-support. Yet, relaxed assumptions, in some cases, become necessary to address more complex real-world situations and to strike a balance between model simplicity and utility. In contrast, tightening the models is a natural move to elevate the quality of solutions.

An important relaxation issue is associated with the complexity of parts in the nesting models. There is an expectation for a future increase in the adoption of real-world geometries, rather than oversimplified regular geometries, including bounding boxes. This oversimplification may result in suboptimal nesting solutions, impacting material utilization and production efficiency. Future research should include the ability to handle irregular geometries more effectively, allowing for a more realistic representation of each part. Nonetheless, this stresses the importance of the model also incorporating advanced non-overlapping constraints, since the interaction between irregular geometries poses a significant modelling challenge.

Incorporating uncertainties is also essential for delivering robust solutions. For example, uncertainty in demand, parts' characteristics, arrival times, and availability of machines. Some quality issues may also be considered, for example, parts being damaged that need to be produced again.

5.2. Solving interrelated problems

One of those sound integrations is the consideration of AM scheduling with the distribution of parts to the final customer. The scarcity of research here indicates that while the integration of nesting and scheduling problems has witnessed substantial attention, the extension to include distribution of parts is an offshoot where further investigations are warranted.

Another key integration that has unfolded, though with subtle expression, is the consideration of part decomposition concerns with nesting. Given the interrelated nature of these decisions that have been typically addressed separately, there is ample room for integration. Going a little further, it would be interesting to extend the decomposition-and-nesting problem to include scheduling considerations. For example, when a part is divided into smaller parts and grouped into builds, these parts are typically manufactured on a single machine. Nonetheless, an extension of this problem would involve manufacturing the parts into different builds that are then distributed across parallel AM machines.

Other decisions can be further considered with nesting and scheduling problems in AM. One of the less frequent but important ones, is the integration of inventory issues, including supply chain management and spare parts management. AM features should be distinguished from traditional inventory topics. In traditional manufacturing, inventory management often focuses on maintaining stock levels of finished goods, raw materials, and components. Yet, in AM the dynamic nature of production and on-demand manufacturing necessitate a reevaluation of inventory strategies. Integrating inventory decisions into the nesting and scheduling problems in AM facilitates a more streamlined process.

5.3. Modeling emerging AM technologies and scenarios

CMfg and mobile printing robots seem to be two key technologies for leveraging the AM system and vice versa. CMfg is a sound avenue to further improve the utilization of idle AM machines and achieve cost reductions based on collaborative production models. Although promising, CMfg has only recently been brought into the discussion of AM scheduling problems, and thus, there is still ample room for further research. For example, the consideration of factors such as dynamic arrival of orders, acceptance of orders, downtime of AM machines, and capabilities of logistic services are all needed to further advance CMfg AM scheduling.

Collaborative multi-robot manufacturing suggests a new paradigm to provide scalability to AM, but as for CMfg AM scheduling research, also the research on AM multi-robot scheduling is still in its infancy. A major challenge is to delineate multiple chunks within complex shapes. In this domain, different avenues to represent the chunks could be considered. Furthermore, existing works utilize a limited number of robots. Future research should focus on the application of more robots for cooperative AM, promoting higher utilization of the available robots.

Another technological improvement in AM systems is multi-material technologies, such as the multi-material SLA machines, which can produce a single part using different materials. This technique could simplify certain decisions, including the scheduling of parts from different materials in the same build. A paper exploring this technology is Kim et al. (2010). Given the surging demand for orders requiring multiple materials, multi-material technologies can be used for diverse applications that can take advantage of not only SLA but many other AM technologies. Investigations could also be extended to specialized AM machines supporting multi-laser, multi-extruder, and multi-axis. Furthermore, advanced AM machines that are capable of a multi-step process can reduce the complexity of production scheduling.

5.4. Integrating AM into existing processes

It is foreseeable that AM and traditional processes will coexist.

Nevertheless, only few works have modelled the AM scheduling problem based on these conditions (Makanda et al., 2022; Kapadia et al., 2022; Rossi and Lanzetta, 2020; Freens et al., 2015). In a coexistence scenario, the decision-maker has the flexibility to select between AM machines and traditional manufacturing machines for producing a set of specific parts. This decision can prove advantageous for planning purposes, considering that AM is characterized by reduced throughput and relatively high costs, even though the specific cost is contingent on factors including material choices and build volume (Maranha et al., 2023). However, in opposition to traditional manufacturing machines, AM enables the production of geometrically complex parts and offers flexibility for product design (Maranha et al., 2023). This scenario provides the opportunity to adapt production scheduling, particularly for dynamic orders.

Even when all parts are manufactured through AM, there is evidence suggesting that, specifically, post-processing stations are required to ensure surface quality, support removal, and dimensional accuracy. However, most AM scheduling studies are based on single and parallel machine settings. This suggests an opportunity to extend the research horizon to incorporate flow-shop and job-shop models. Rework and repair operations can also be included as downstream AM options. Indeed, post-processing operations poses significant challenges for production scheduling and control since routings and processing requirements vary. This may result in unbalanced workloads at downstream AM stations. Future research could explore optimization-based workload control. Integrating post-processing operations with workload control and order acceptance decisions seems to be another interesting research stream.

5.5. Adopting new solution approaches

Our analysis demonstrated that hybrid algorithms that blend meta-heuristics with machine learning techniques represent the most recent advancements in solution approaches. More research is required to further advance this promising field. Meanwhile, machine learning is widely used to recognize patterns, which is commonly referred to as computer vision. In AM, for part orientation decisions, machine learning could analyze the characteristics of 3D CAD models and determine the optimal orientation for each part. This approach can minimize support structures, reduce build time, and improve the quality of parts. Additionally, for part placement, machine learning can autonomously arrange parts on the build. Moreover, a distinct aspect is that these algorithms can learn from historical data and adapt to different shapes and product requirements.

Another opportunity lies in the application of machine learning to dynamic manufacturing, based on real-time factors, including machine and material availability, and machine downtimes. Additionally, these models can be trained to predict the quality of parts, considering various parameters, including machine settings, material characteristics, and environmental conditions. Another potential venue is the use of machine learning to optimize AM supply chain processes, including inventory management and logistics.

5.6. Achieving environmental sustainability

A paramount concern pervading several manufacturing systems to-date relates to the environment and the sustainability of operations. By examining various aspects of AM, from material selection to printing techniques, researchers have identified opportunities to minimize the carbon footprint of AM operations, specifically studying the reduction of energy consumption and cost through nesting and scheduling or adjusting of process parameters. For example, Karimi et al. (2021) conducted a first study addressing energy issues while scheduling AM machines, while Tafakkori et al. (2022) and Lin et al. (2023) focused on reducing energy consumption by better utilizing the build space of machines and scheduling. Rupp et al. (2021) and Kucukkoc (2023) were

among the first to address the AM machine scheduling considering emissions caused by production and transportation activities.

Despite these promising first steps, future research is needed that expands and deepens the investigation of carbon emissions in AM. This includes modeling low-emission strategies in which nesting and scheduling decisions benefit, for example, from better material usage, lower energy consumption, and more sustainable supply chains. In particular, additional research work is required to reduce material waste (some AM processes may generate more waste material than others) and remanufacturing possibilities (this could involve fixing damaged parts or making design changes to enhance the durability or functionality of parts). Regarding energy consumption, more work can be done to improve energy efficiency and reduce the energy bill in AM. Incorporating renewable energy sources is a new challenge that may require considerations of the intermittency and variability of renewable energy generation and how it impacts the overall energy consumption profile of AM technologies. Finally, future work should address the carbon emissions associated with production and transportation in AM supply chains. This involves production scheduling decisions aimed at reducing peak power demand and shift electricity usage in response to time-varying rates, along with optimizing transportation modes and distance from raw materials to manufacturing and from manufacturing to the customer.

6. Conclusion

This paper set out to summarize the challenges posed by AM in the field of operational research. Science mapping techniques and traditional systematic review methods were used to capture the progression of knowledge on nesting and scheduling models in the context of AM. One major critique on the proposed models is that most of them still lag behind the practical needs of real manufacturing systems. In the midst of this issue, industries are demanding solutions to smoothly integrate AM processes into their supply chains and manufacturing systems.

As main findings of our study, the nesting and scheduling problems can be classified into six decision types. Modeling the interdependencies between all these decisions poses a complex, yet important problems for research. Although we notice that there are various assumptions that restrict existing models, we argue that nesting and scheduling research in AM has experienced notable advancement in recent years. Our findings demonstrate that this progression is supported by sound research streams that include new perspectives and functional integrations to the primary problem. In short, we identified relevant research problems: part decomposition, part distribution, order acceptance and scheduling, processing operations, collaborative manufacturing, and multi-robot manufacturing to be addressed together with the nesting and scheduling problem.

CRedit authorship contribution statement

Marcelo Pinto: Conceptualization, Data curation, Writing – original draft, Writing – review & editing, Investigation. **Cristóvão Silva:** Conceptualization, Investigation, Methodology, Validation, Resources. **Matthias Thürer:** Writing – review & editing, Methodology, Validation. **Samuel Moniz:** Conceptualization, Funding acquisition, Writing – original draft, Writing – review & editing, Investigation, Methodology, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that support the findings of this study are available from the

corresponding author, Samuel Moniz (samuel.moniz@dem.uc.pt), upon reasonable request.

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